

9(a)	energy required to separate (all) the nucleons (in the nucleus)	M1
	to infinity	A1
9(b)(i)	$\Delta m = \{[(84 \times 1.007276) + (128 \times 1.008665)] - 211.942749\} \text{ (u)}$ $(= 1.778 \text{ u})$	C1
	$= 1.778 \times 1.66 \times 10^{-27} \text{ (kg)}$	C1
	$= 2.95 \times 10^{-27} \text{ kg}$	A1
9(b)(ii)	$E = (\Delta)mc^2$	C1
	binding energy $= 2.95 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 2.66 \times 10^{-10} \text{ J}$	A1
9(b)(iii)	binding energy per nucleon $= (2.66 \times 10^{-10}) / 212$ $= 1.25 \times 10^{-12} \text{ J}$	A1
9(c)(i)	line rising to a single peak that is to the left of the '9' in the Fig. 9.1 label and then continually decreasing	B1
	steep positive gradient on the left of the peak and shallow negative gradient on the right	B1
9(c)(ii)	X shown on the line at a value of A that is to the right of the left-hand edge of the 'A' in the axis label, and to the left of '2' in the 250 label	B1
9(c)(iii)	nucleus formed (as a result of the decay) has a lower nucleon number	B1
	(nucleus formed has a) greater binding energy per nucleon	B1